

National University of Computer and Emerging Sciences



Fundamentals of Software Engineering

Poet's Sanctuary: Urdu and Punjabi Poetry Platform.

Sheher-e-Sukhan

Project Title and Introduction

Project Title: Sheher-e-Sukhan – A Platform for Exploring Urdu and Punjabi Poets

Introduction:

Sheher-e-Sukhan is a desktop-based application designed to provide users with an interactive platform to explore and engage with Urdu and Punjabi poetry. Developed using C# and SQL, the application allows users to create accounts, save their favorite poems, manage personal preferences, and discover poets based on their interests. Administrators have access to managing the poetry database and monitoring user engagement.

User Stories

Project Setup – As a developer, I want to set up the *Sheher-e-Sukhan* project so that I can start working on the application.

Data Models Implementation – As a developer, I want to create the main database structure so that the app can store and manage data properly.

User Registration with Preferences – As a new user, I want to sign up and select my preferred poets so that I get a personalized poetry experience.

User Login & Authentication – As a user, I want to log in securely so that I can access my saved poems and preferences.

Admin Login & Authentication – As an admin, I want to log in securely so that I can manage poetry content and monitor user engagement.

User Dashboard Access – As a user, I want to have a personal dashboard where I can view my saved poems, preferences, and recent activity.

Admin Dashboard Access – As an admin, I want to have an admin dashboard where I can manage users, review activity, and update poetry content.

Main Page Access – As a user, I want to access the main page after logging in so that I can explore poetry content and navigate to different sections eas1. Poet Search

Poetry Submission –As a user, I want to submit my own poetry so that others can read and comment on it.

Poetry Recitation – As a user, I want to listen to poetry recitations so that I can enjoy poetry in an audio format.

Browse by Theme – As a user, I want to browse poetry by its theme (e.g., love, nature, philosophy) so that I can explore poems that interest me.

View Poems by Theme – As a user, I want to view a collection of poems categorized under a specific theme, so I can explore them all at once.

Poet Follow Feature – As a user, I want to follow poets so that I can stay updated on their work.

Unfollow a Poet – As a user, I want to unfollow a poet if I no longer wish to receive updates from them.

Participate in Poetry Contests – As a user, I want to participate in poetry contests so that I can showcase my talent.

Vote on Contest Entries – As a user, I want to vote on contest entries so that I can support my favorite poems.

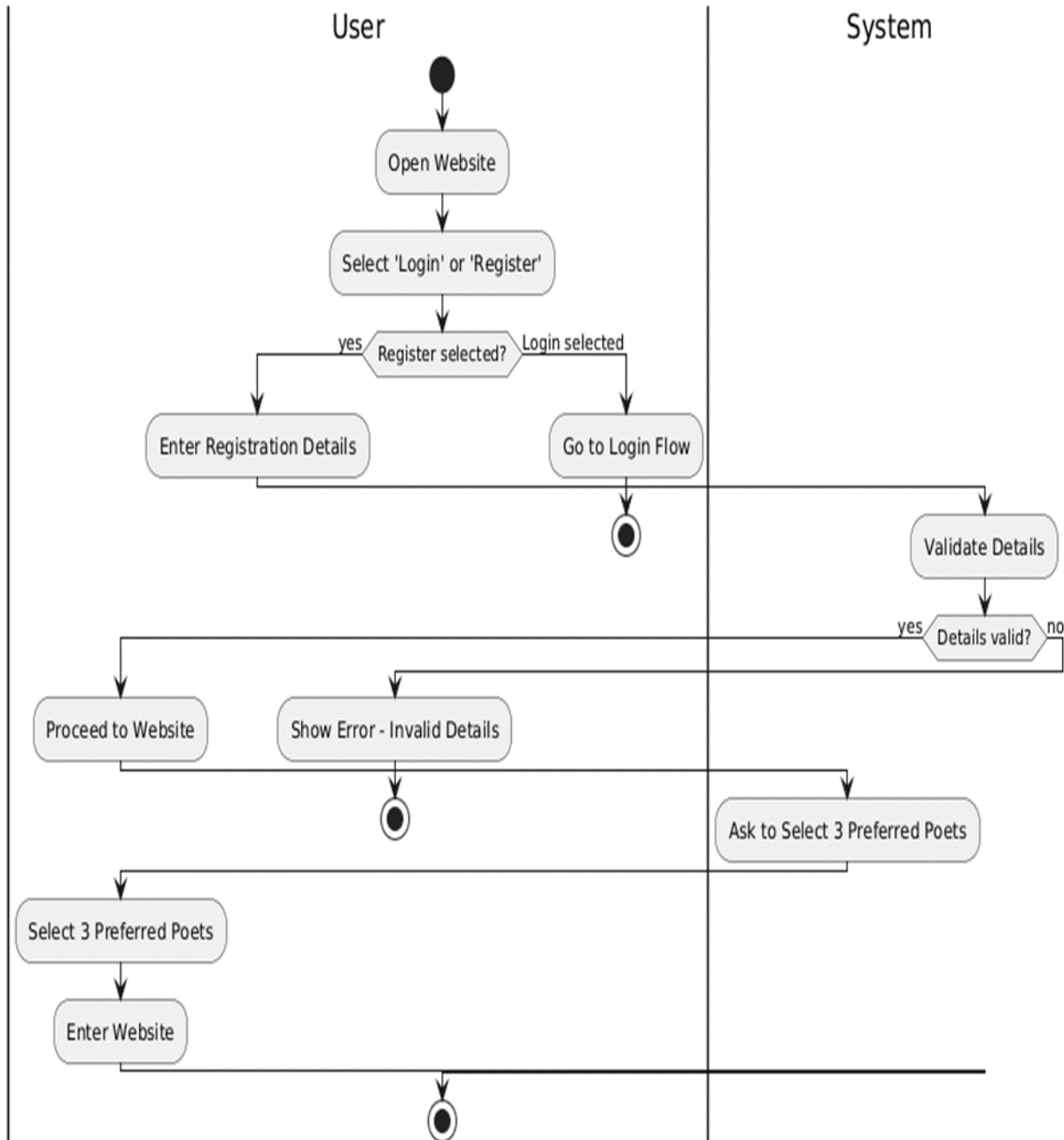
Monitor Poetry Analytics – As an admin, I want to monitor poetry analytics so that I can track user engagement.

View Poet Timelines – As a user, I want to view poet timelines so that I can learn about their literary journey.

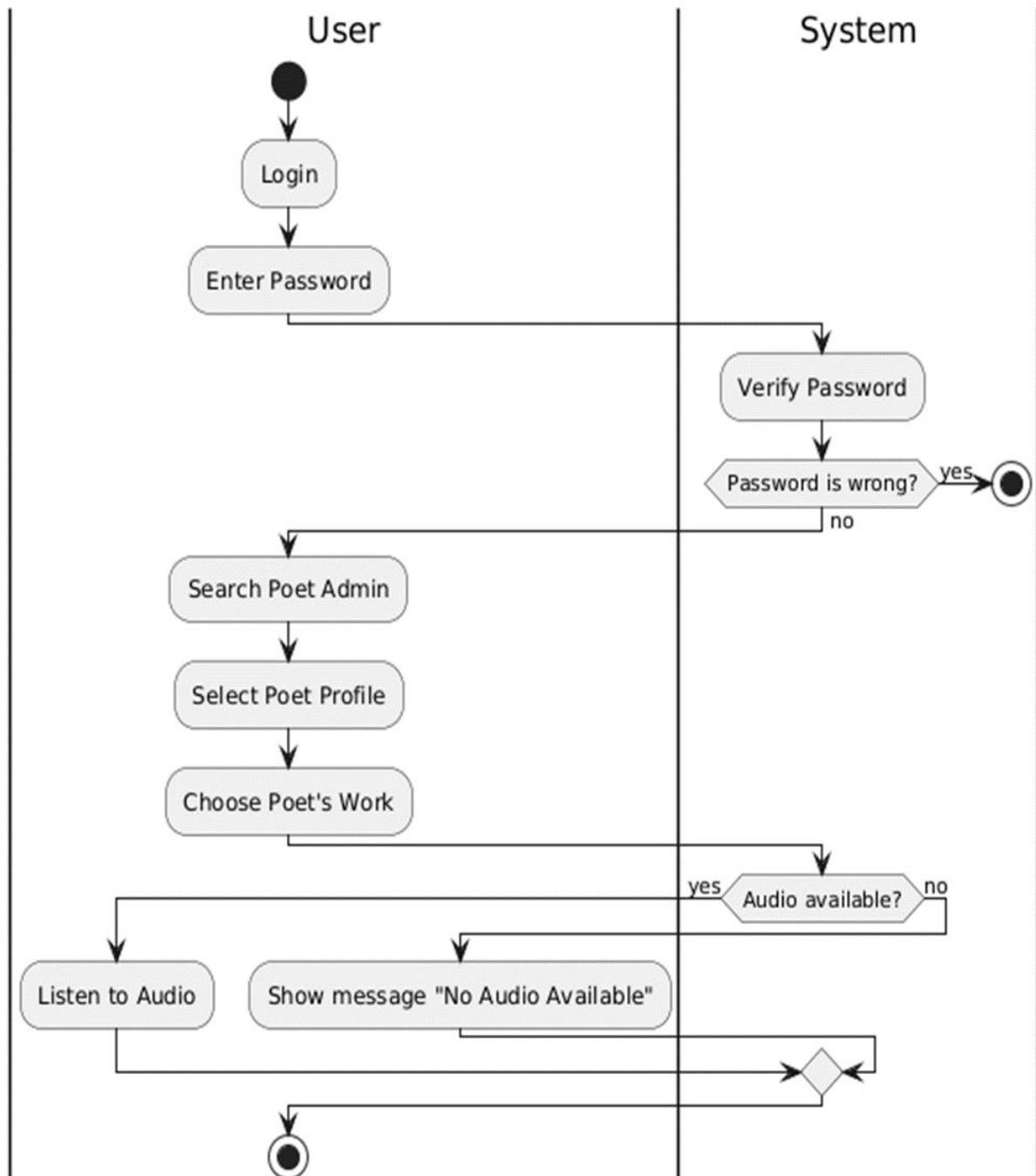
Track My Activity History – As a user, I want to track my activity history so that I can review my contributions.

Design:

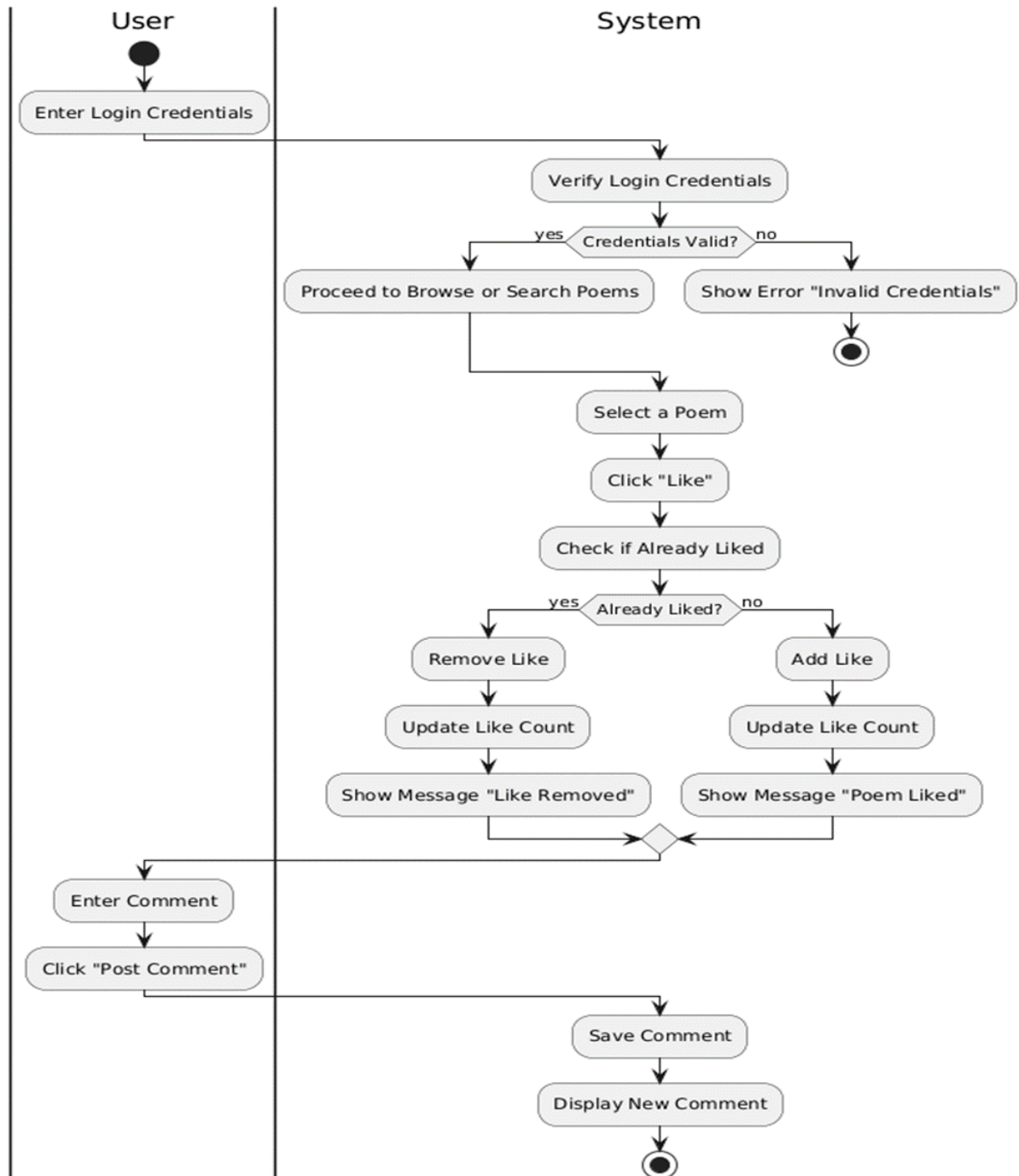
Activity Diagram (Registration of User):



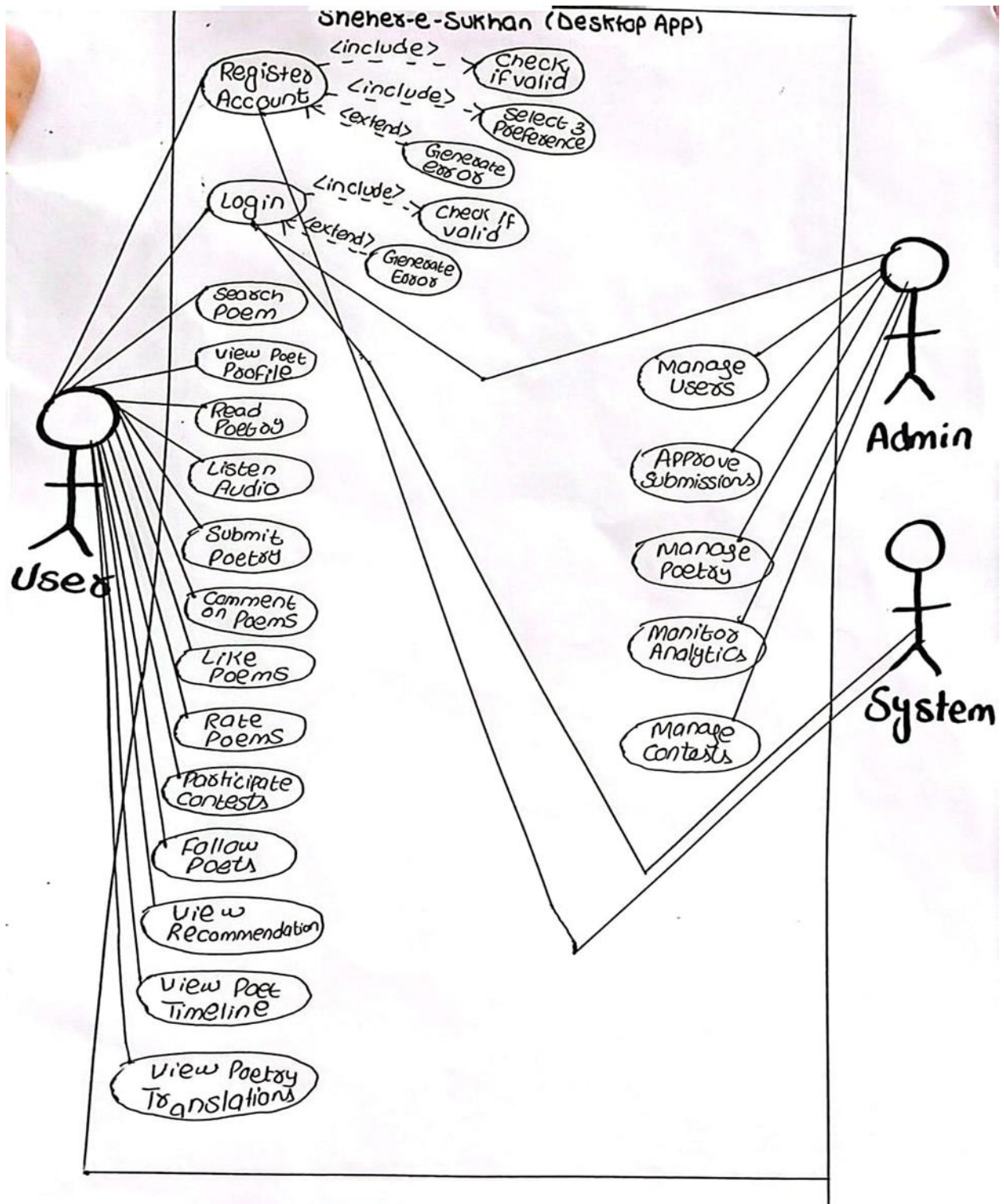
Activity Diagram (Listen To Poetry Audio):



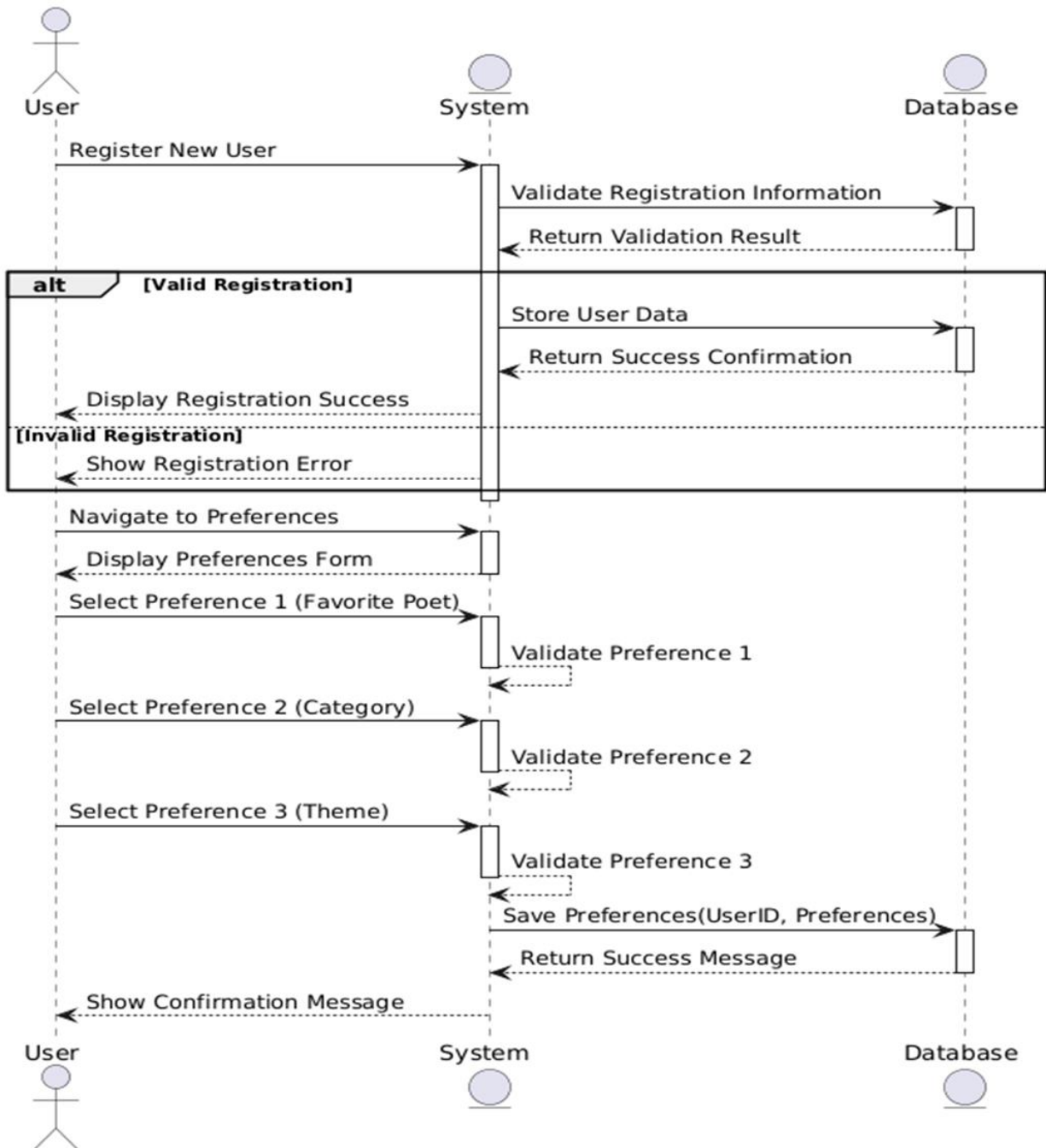
Activity Diagram (Comment and Like Poetry):



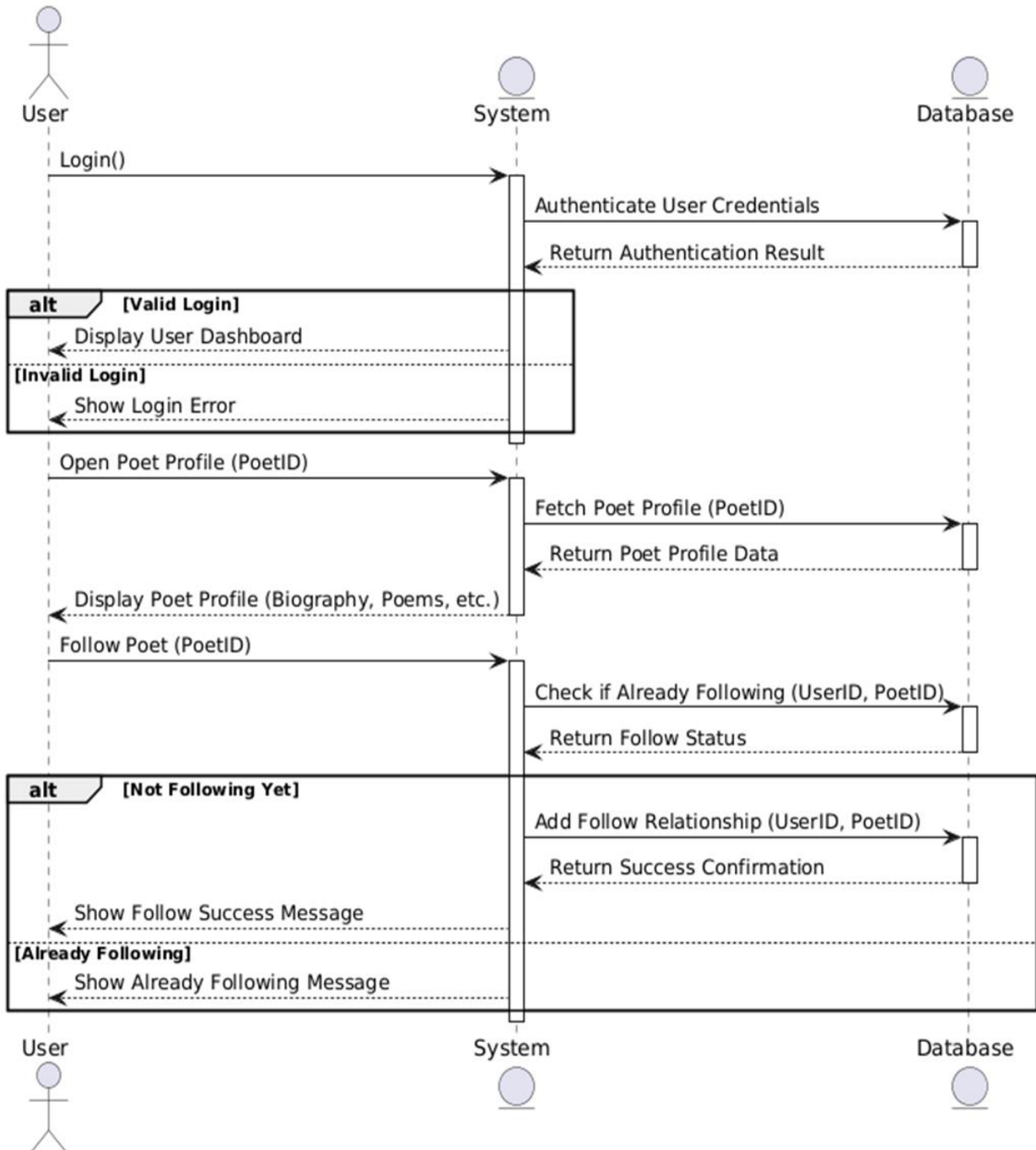
Use Case Diagram:



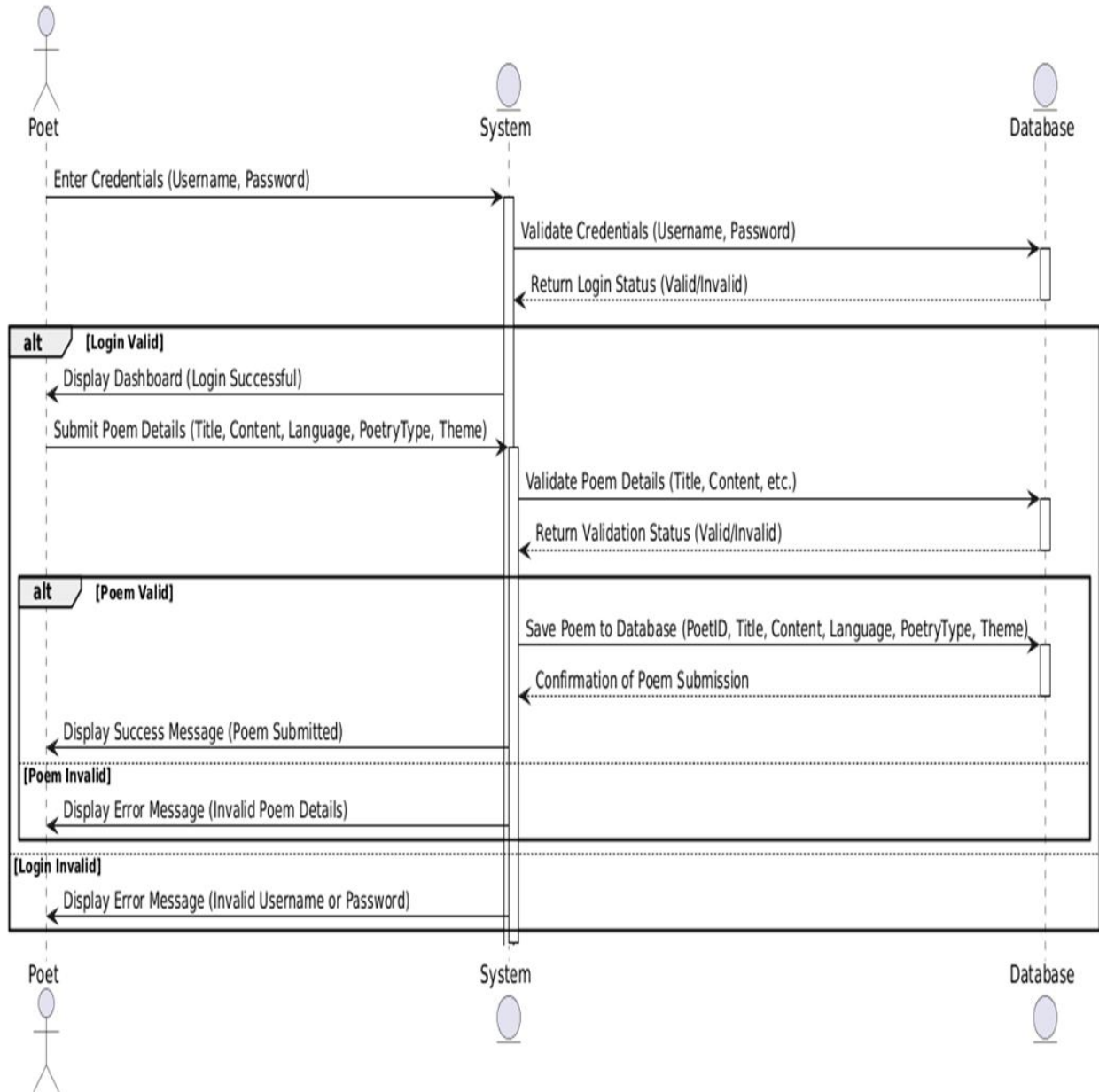
Sequence Diagram (Registration and Selecting 3 Preferred poets):



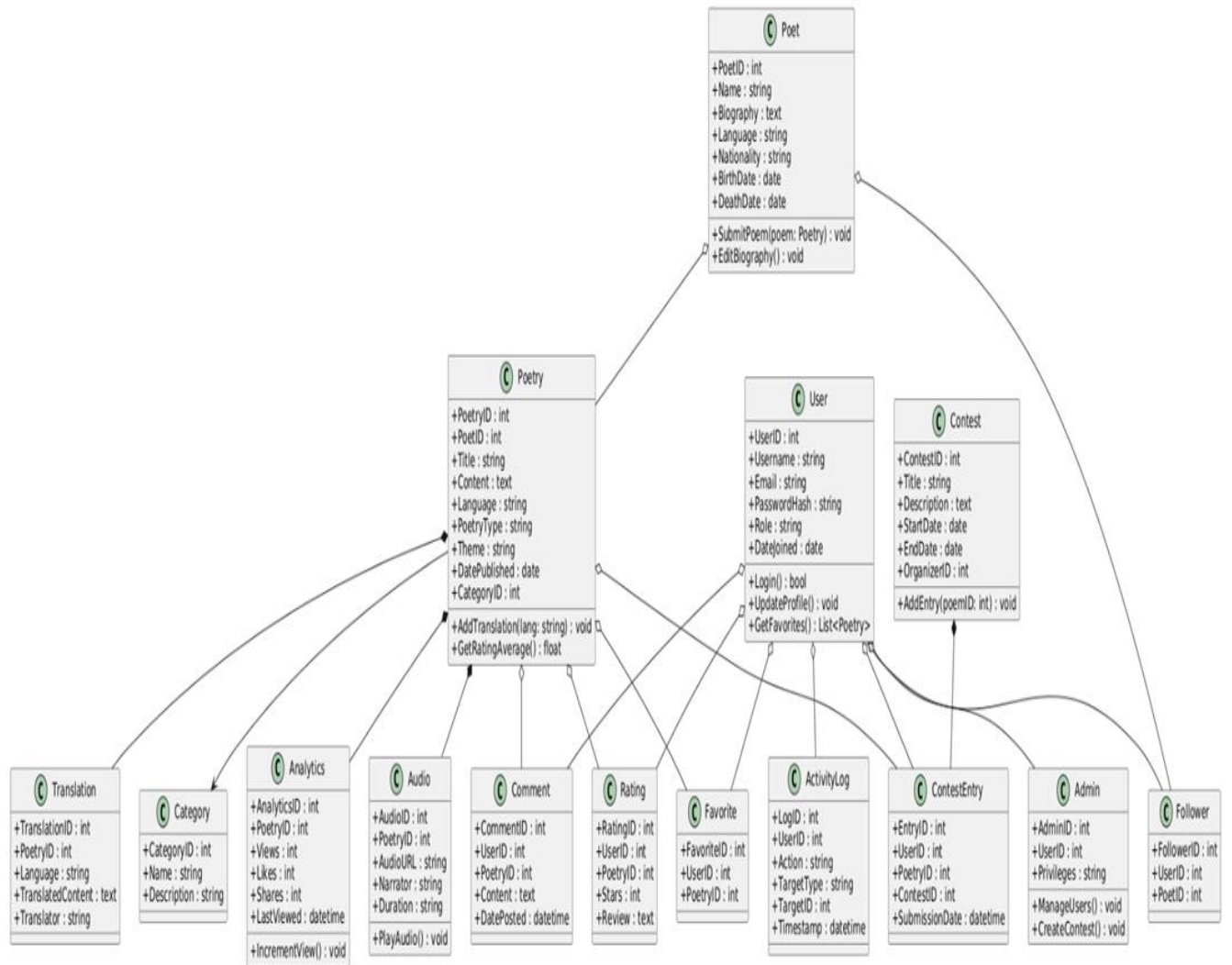
Sequence Diagram (Follow Poet):



Sequence Diagram (Submit a Poetry):

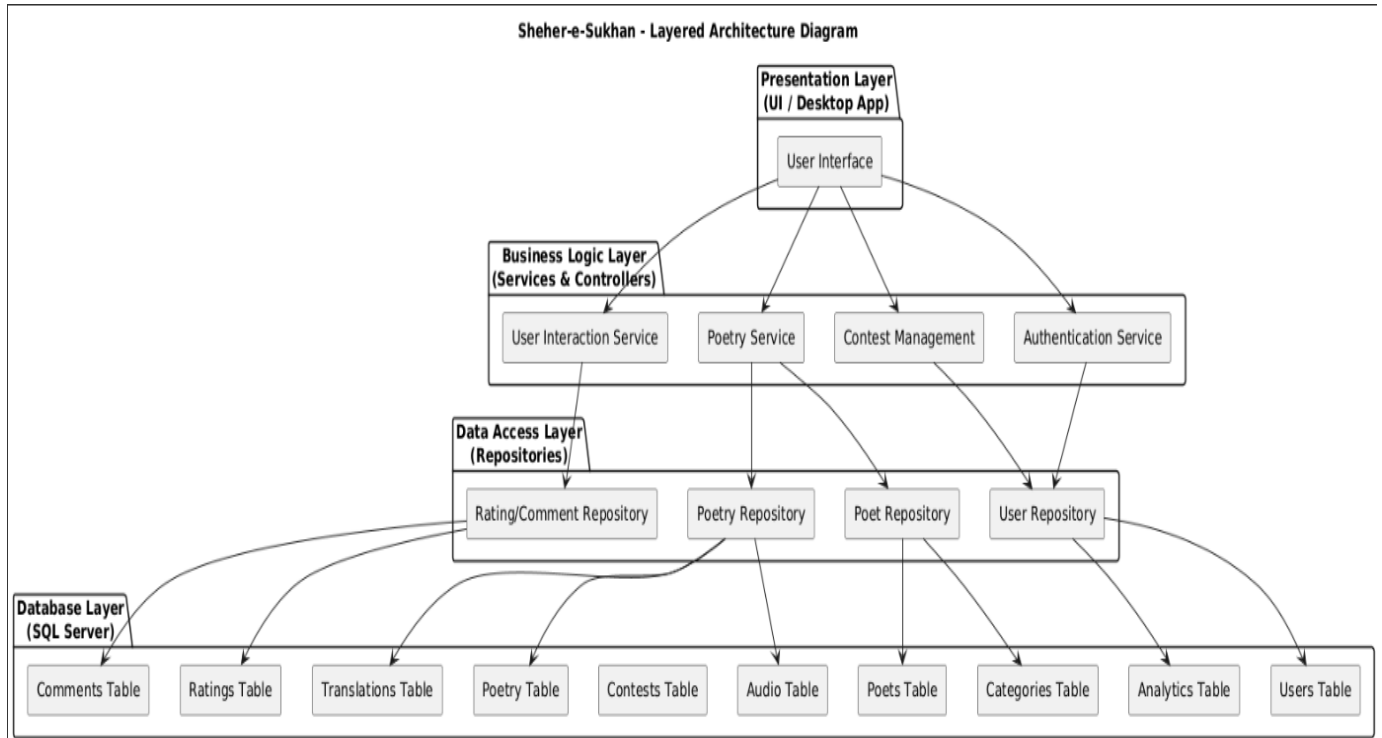


Class Diagram:



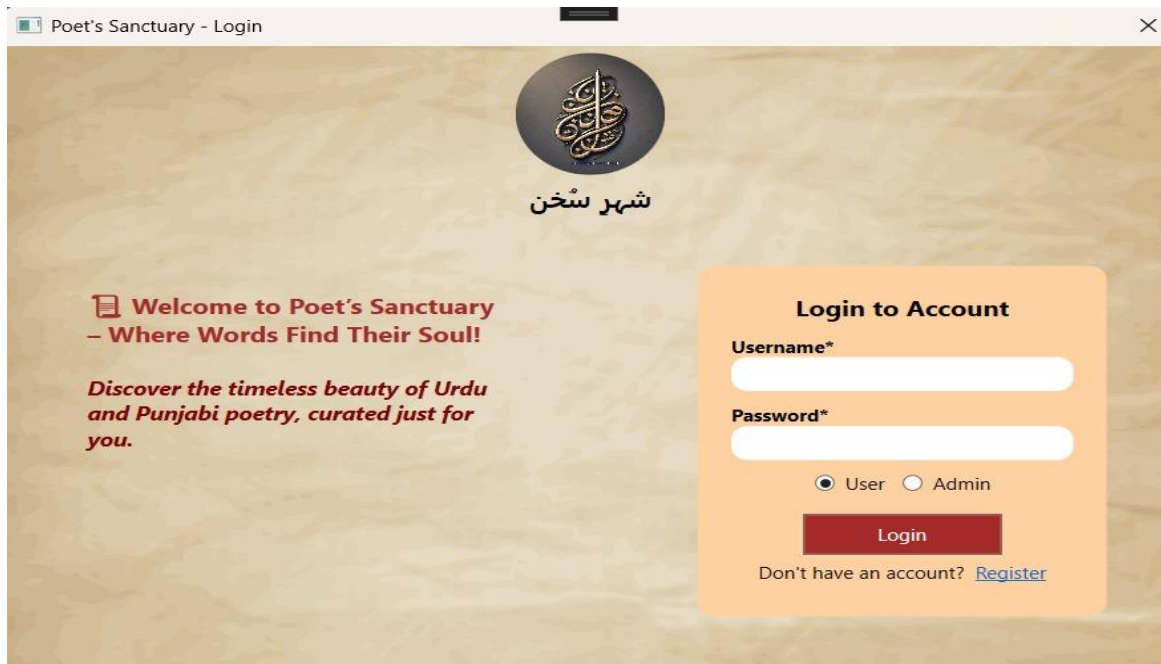
Architecture:

For the Sheher-e-Sukhan desktop application, an ideal architecture pattern is the **Layered Architecture**. It separates concerns into distinct layers, enhancing maintainability, scalability, and testability.



Actual Implementation ScreenShots:

Login page:



Poet's Sanctuary - Login

شہرِ سخن

Welcome to Poet's Sanctuary
– Where Words Find Their Soul!

Discover the timeless beauty of Urdu and Punjabi poetry, curated just for you.

Login to Account

Username*

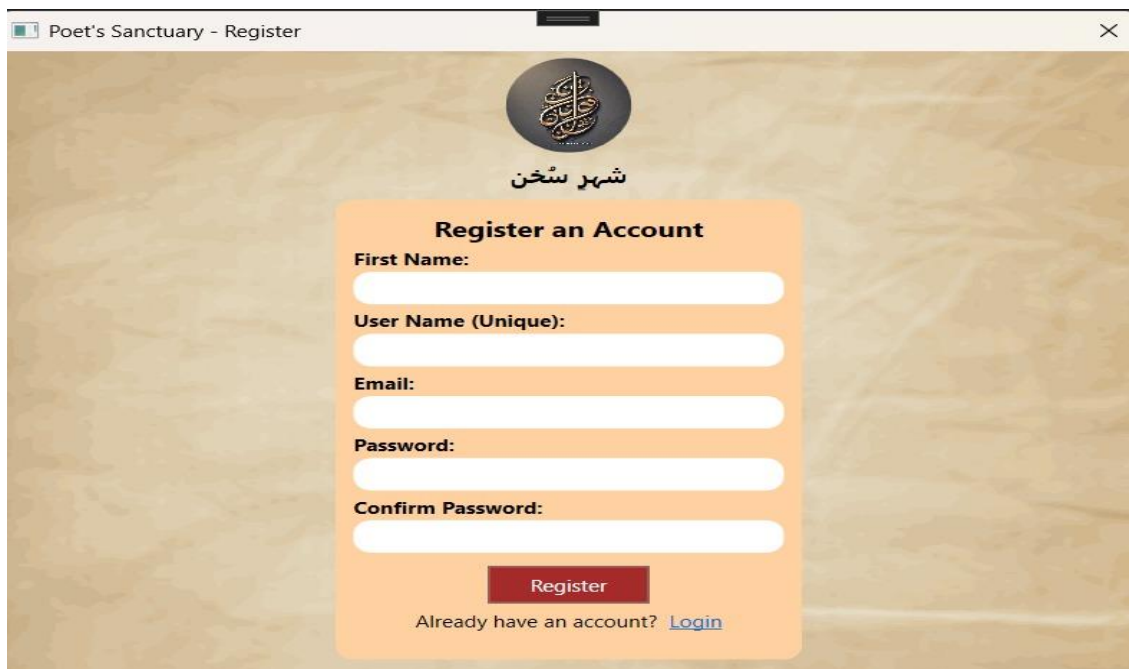
Password*

☒ User ☐ Admin

Login

Don't have an account? [Register](#)

Registration page:



Poet's Sanctuary - Register

شہرِ سخن

Register an Account

First Name:

User Name (Unique):

Email:

Password:

Confirm Password:

Register

Already have an account? [Login](#)

Select Preference after registration:

Select Preferences

شہرِ سخن

Select 3 Preferred Poets

☐ Allama Iqbal

☐ Faiz Ahmed Faiz

☐ Mirza Ghalib

☐ Ahmed Faraz

☐ Proceed

Main Page:

Home

Poets About Contact Us

شہرِ سخن

Mirza Ghalib

Faiz Ahmad Faiz

Jaun Elia

Mir Taqi Mir

Ahmad Faraz

Search Poets by Category

• Nazm

• Ghazals

• Qasida

• Hamd

• Naat

Select Language

☒ Urdu

☐ Punjabi

Search Poets by Theme

Philosophy

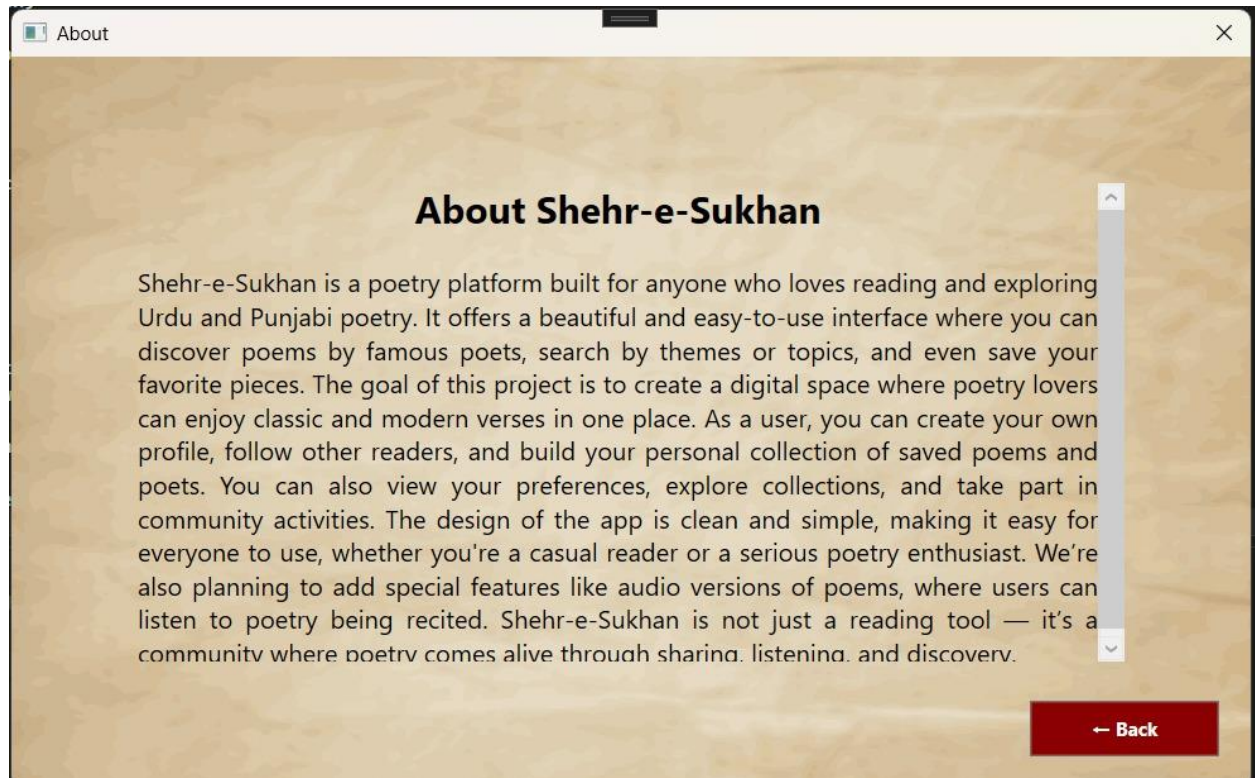
Patriotism

Love

Sadness

Hopelessness

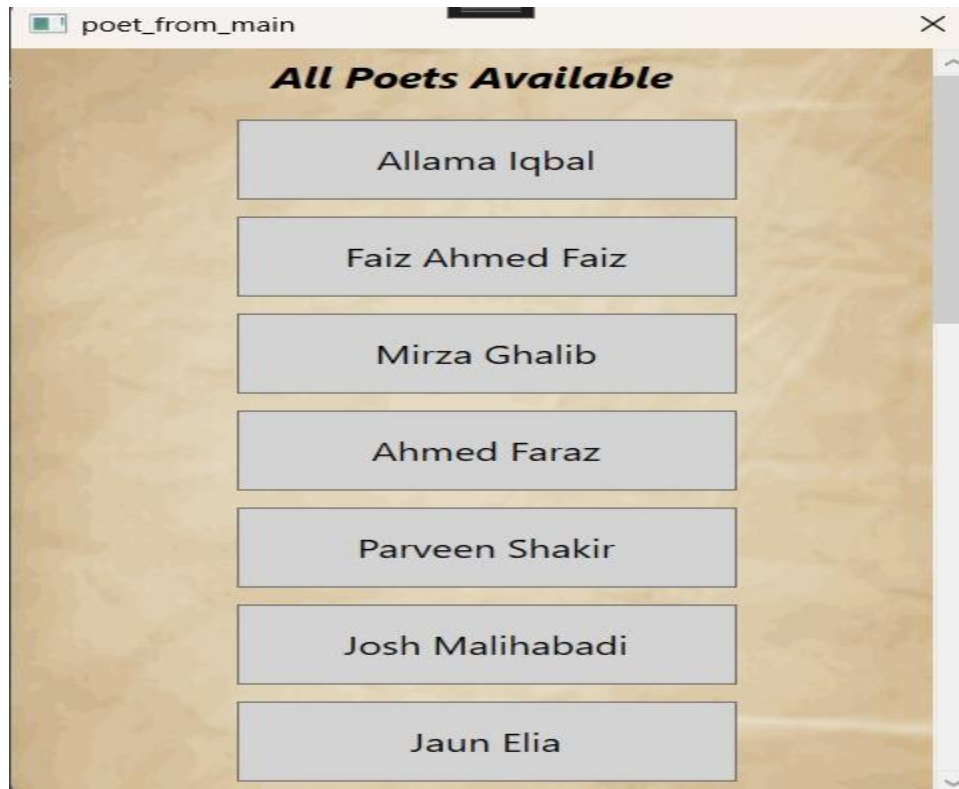
About Sheher e Sukhan:



Selected a specific theme:



All poets:



Poet profile:



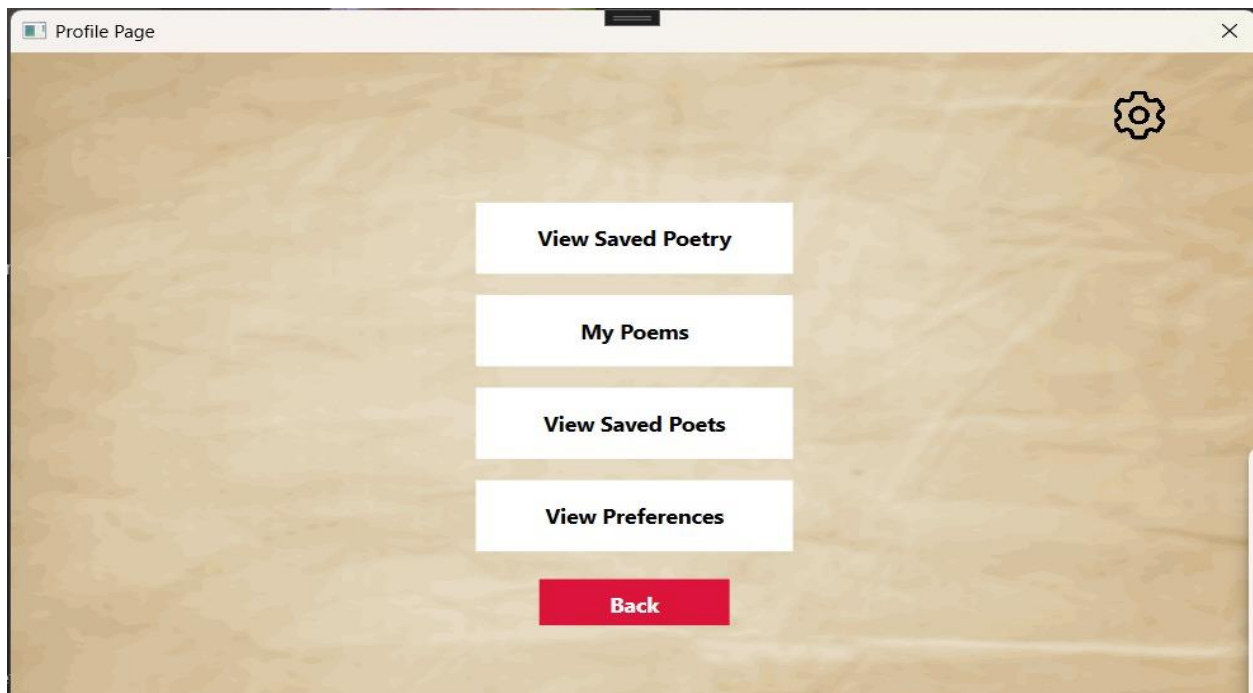
Viewed a Poetry:



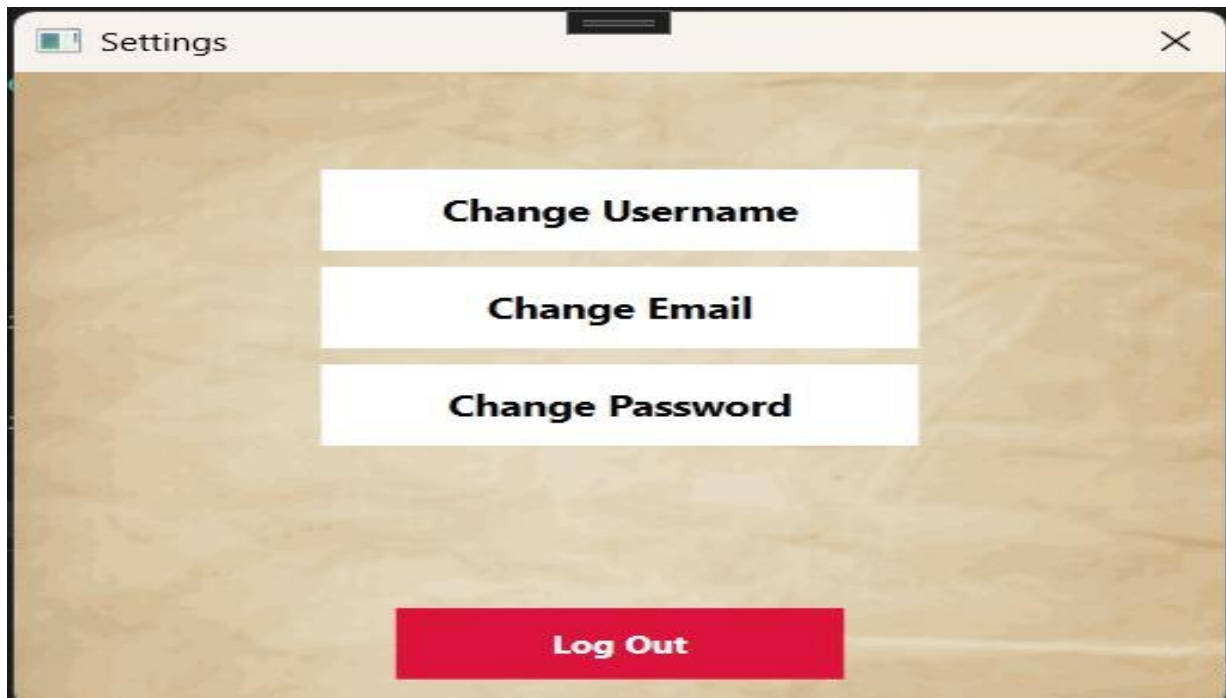
User Profile View:



For User:



Settings:



Admin Dashboard:

Admin Dashboard

Load Users
Load Admins
Announce Contest

User Information

Full Name	Username	Email

First Name:

Username:

Password:

Announce Poetry Contest (ADMIN)

Announce Poetry Contest

Title:

Theme:

Start Date:

Select a date

15

End Date:

Select a date

15

Announce Contest

Poetry Submission Form:

Submit Your Poetry

Poetry Title:

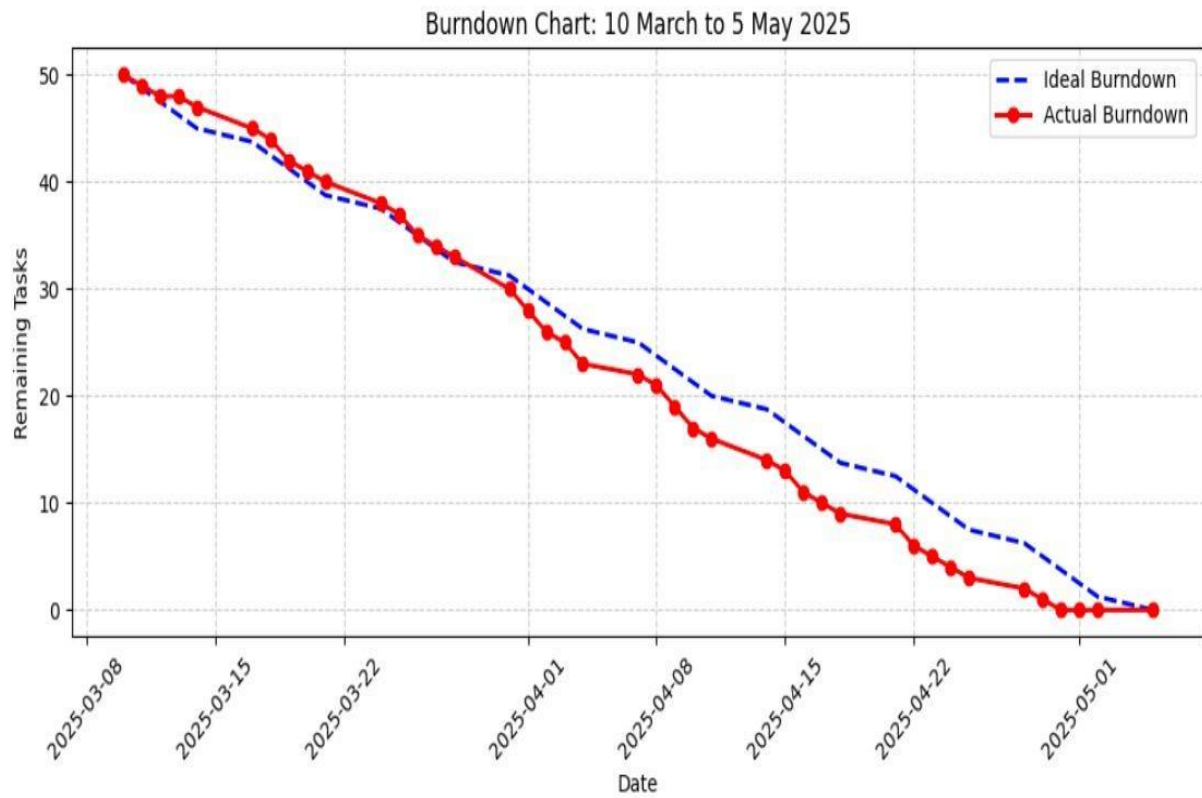
Poetry Text:

Genre:

Theme:

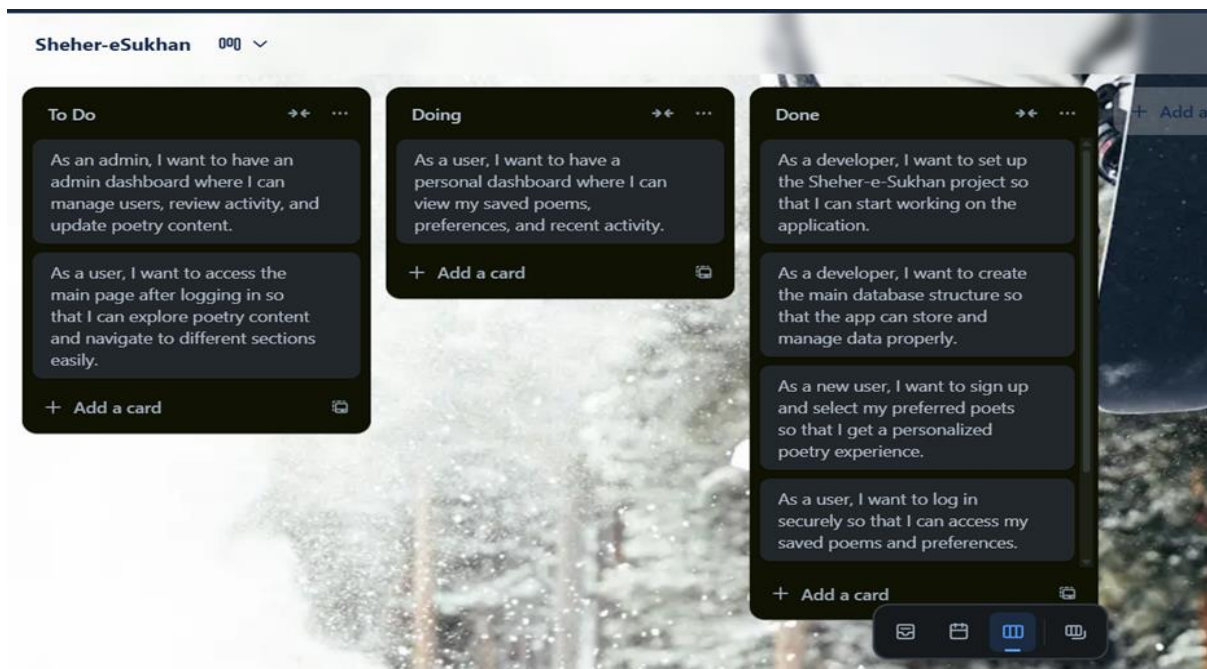
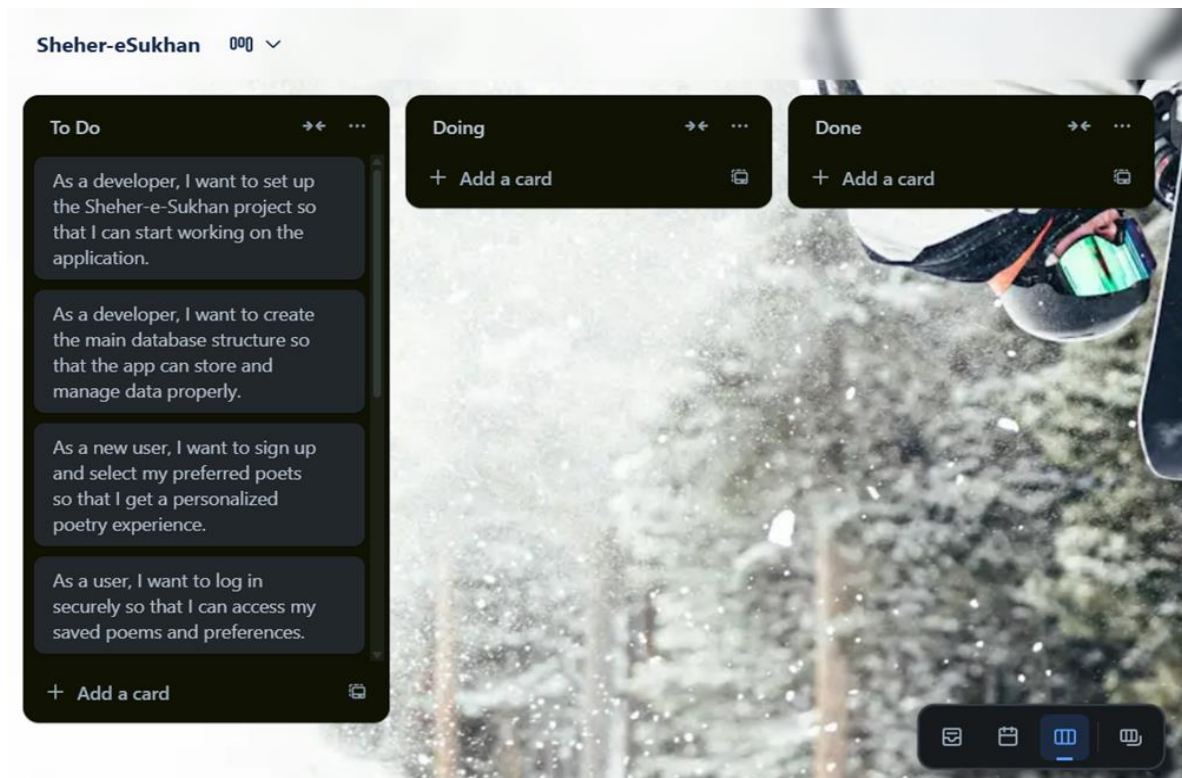
Submit Poetry

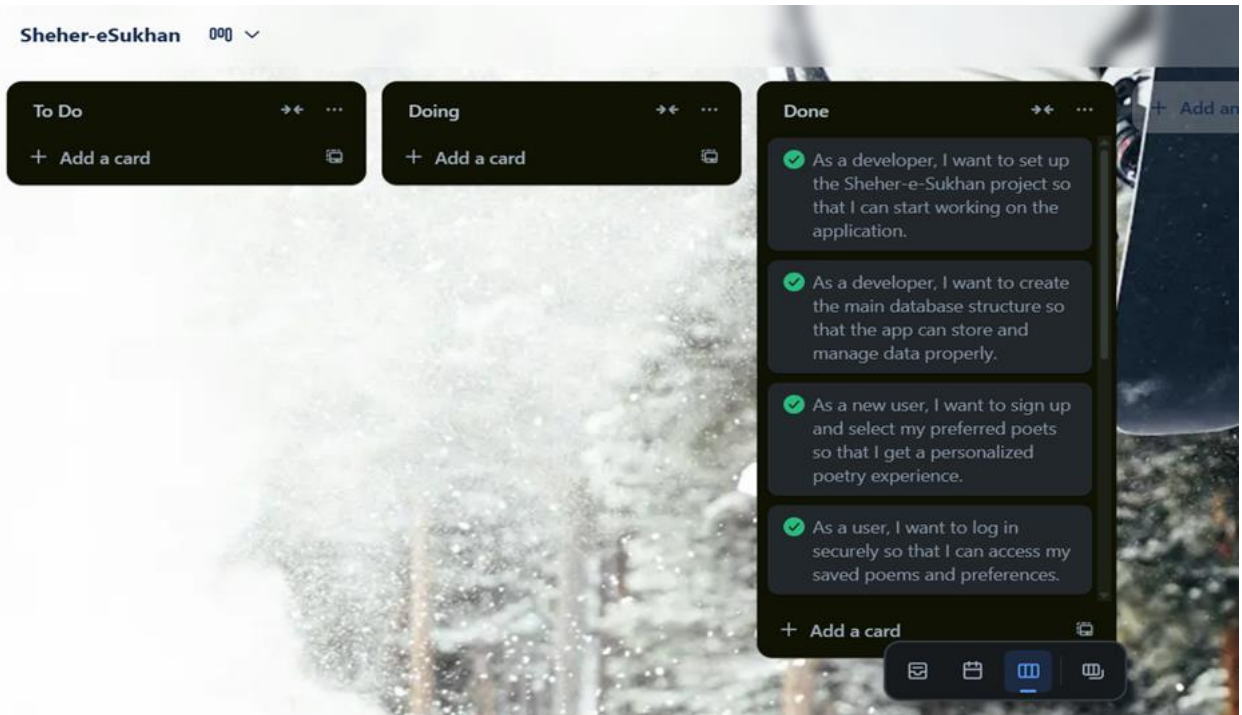
Product Burndown Chart:



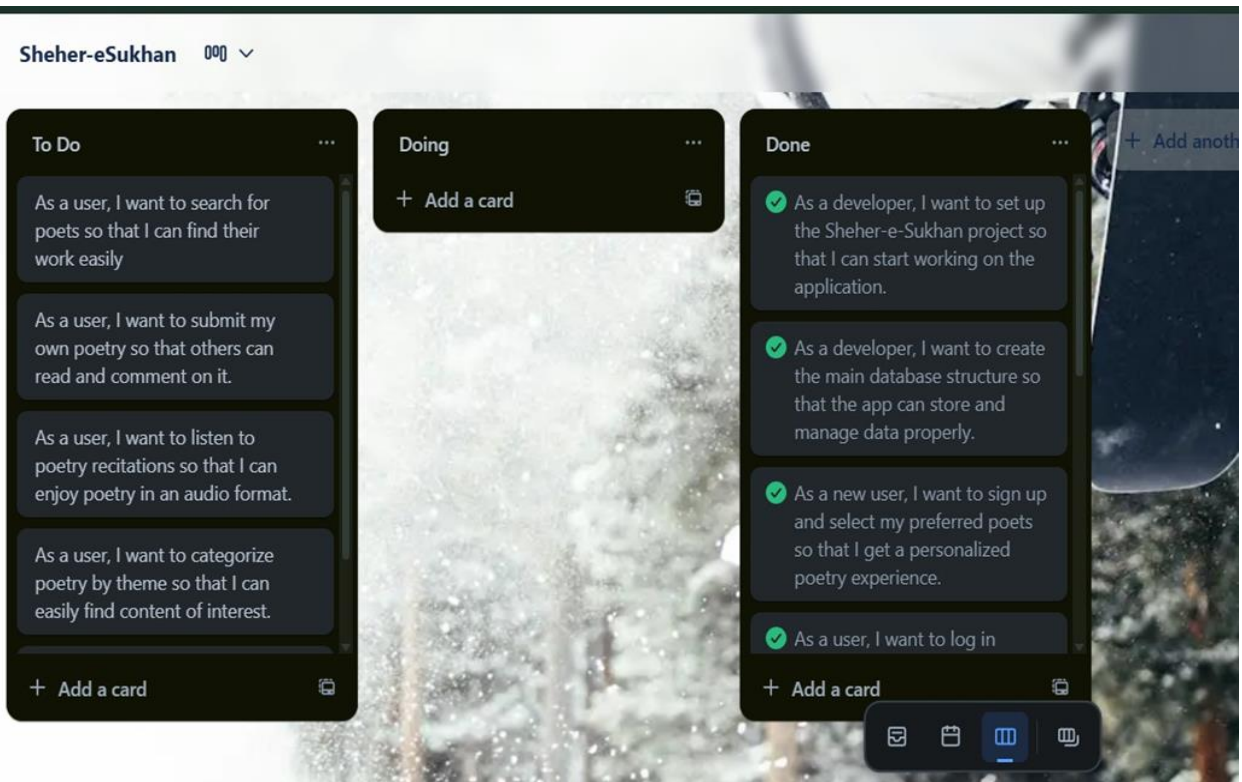
Trello Board ScreenShots:

Sprint 1: Planned: 8 Done: 8





Sprint 2: Planned: 5 Done: 5



To Do

As a user, I want to listen to poetry recitations so that I can enjoy poetry in an audio format.

+ Add a card

Doing

As a user, I want to submit my own poetry so that others can read and comment on it.

As a user, I want to categorize poetry by theme so that I can easily find content of interest.

+ Add a card

Done

that I can explore poetry content and navigate to different sections easily.

✓ As an admin, I want to have an admin dashboard where I can manage users, review activity, and update poetry content.

✓ As a user, I want to search for poets so that I can find their work easily

✓ As a user, I want to follow poets so that I can stay updated on their latest work.

+ Add a card

+ Add another list

To Do

+ Add a card

Doing

+ Add a card

Done

✓ As a user, I want to search for poets so that I can find their work easily

✓ As a user, I want to follow poets so that I can stay updated on their latest work.

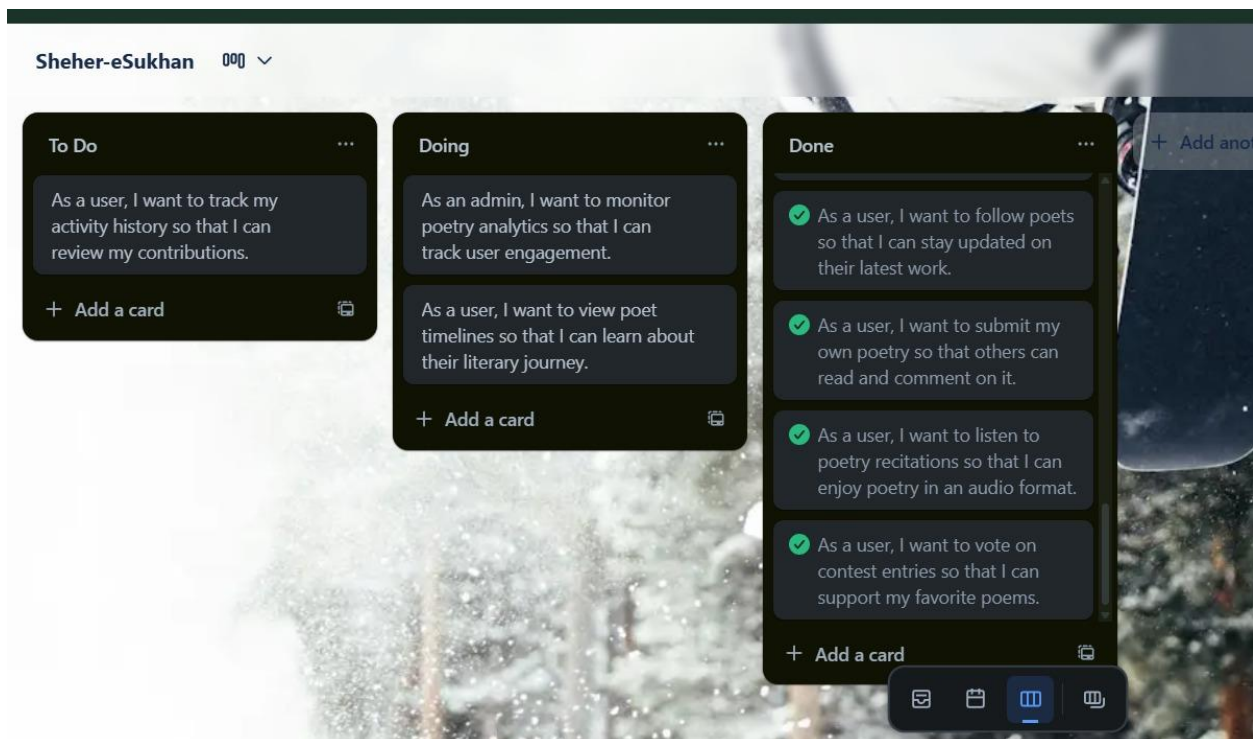
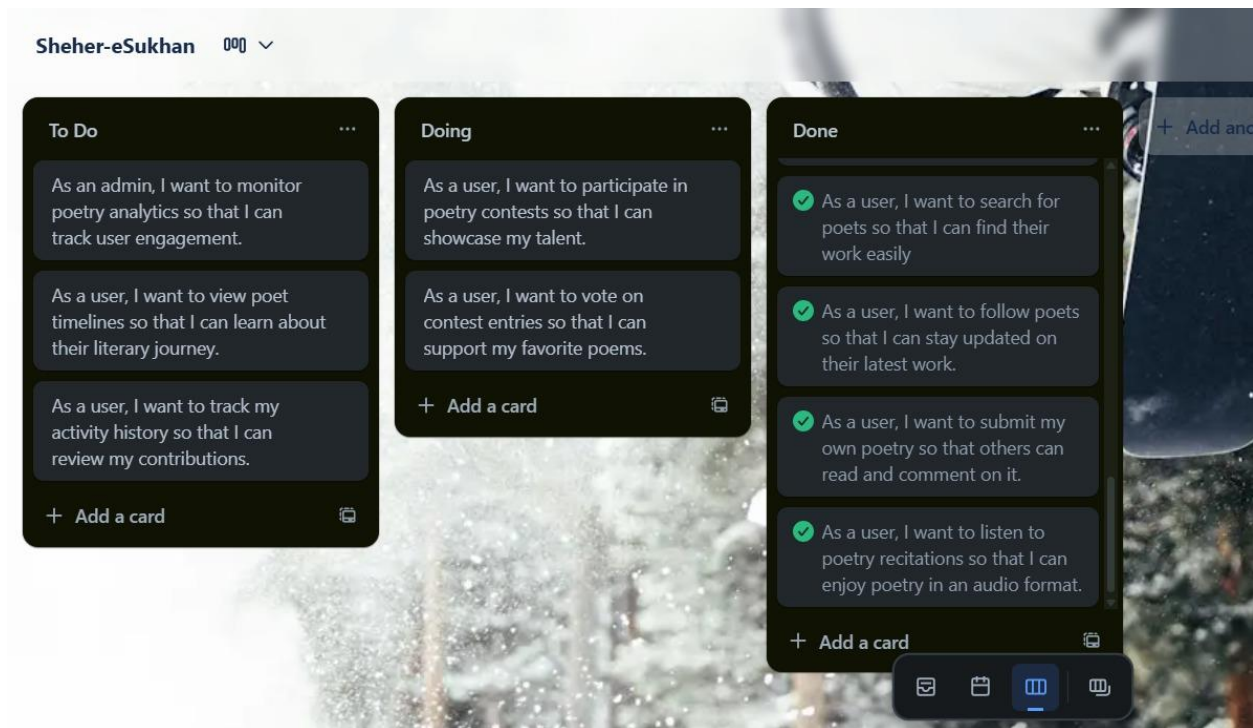
✓ As a user, I want to submit my own poetry so that others can read and comment on it.

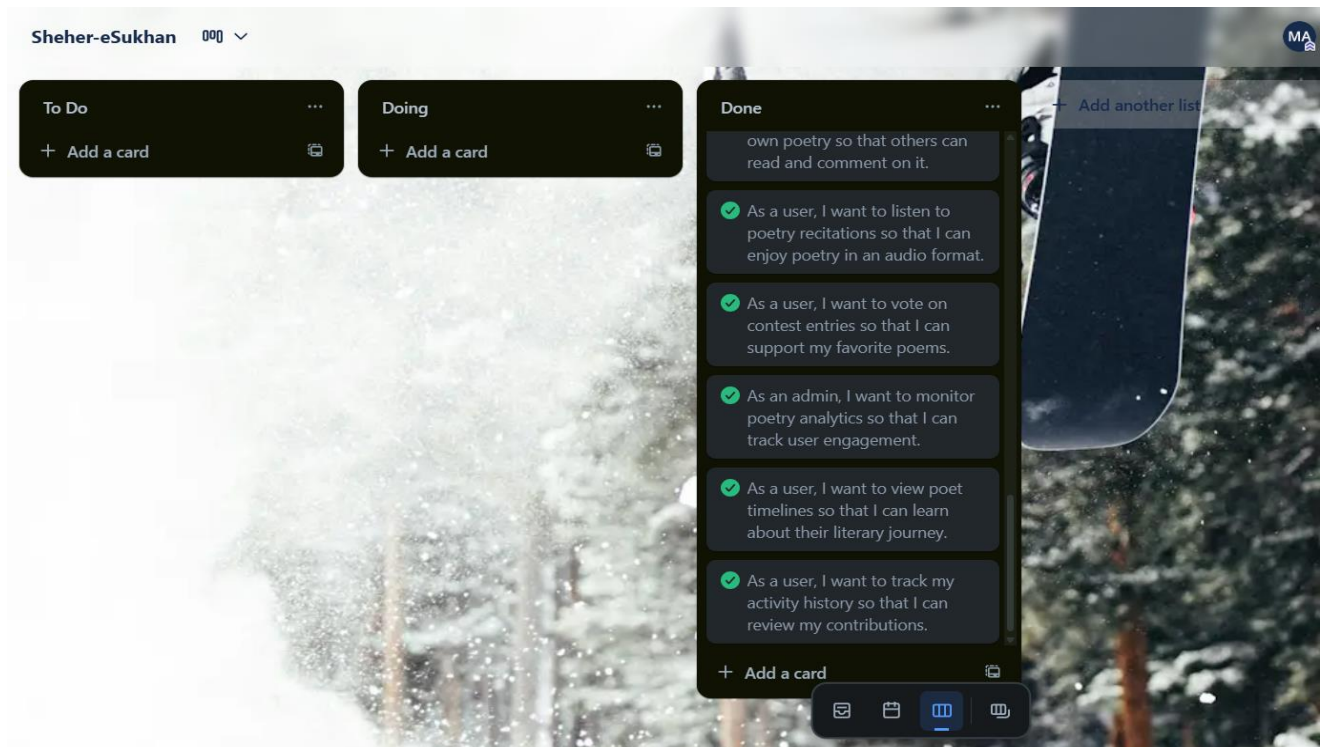
✓ As a user, I want to listen to poetry recitations so that I can enjoy poetry in an audio format.

+ Add a card

+ Add another list

Sprint 3: Planned: 5 Done: 5





Boundary Value Analysis Testing:

Sign up Page:

Key Input Fields

1. **Username:** 2–25 characters, must be unique
2. **Email:** Must be valid format, only Gmail allowed, min length 11
3. **Password:** 6–12 characters, must match confirmation
4. **Confirm Password:** Must exactly match password
5. **Name:** 2–50 characters, cannot be empty

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Field	Range	Valid Range Description	Boundary Values Testing	Expected Outcome
Username	2–25 characters	Must be between 2 and 25 characters, unique	2, 3, 24, 25, 15	1 & 26: ✗Fail 2–25: ✓Pass
Email	≥11 characters, ends in @gmail.com	Must be valid Gmail format and ≥11 chars	"a@gmail.co", "abc@gmail.com", "	Invalid domain/<11: ✗ Valid Gmail ≥11: ✓
Password	6–12 characters	Must be between 6 and 12 characters	6, 7, 11, 12, 8	5 & 13: ✗Fail 6–12: ✓Pass
Confirm Password	Must match Password	Must be exactly the same as Password entered	"abc" vs "abc", "abc" vs "xyz"	Match: ✓ Mismatch: ✗
Name	2–50 characters	Must be between 2 and 50 characters	2, 3, 49, 50, 25	1 & 51: ✗Fail 2–50: ✓Pass

Test Case ID	UserName	Email	Password	Confirm Password	Name	Expected Outcome
TC01	(blank)	valid@mail.com	123456	123456	Amna	✗Reject: Username blank
TC02	ab	ali@gmail.com	123456	123456	Ali	✓Accept
TC03	minahilazeemaz eemminahil	minahil@gmail.com	123456	123456	Minahil	✗Reject: Username too long
TC04	MinahilAzeem	@gmail.com	123456	123456	Minahil	✗Reject: Invalid email
TC05	AhmedBaig	ahmed@gmail.com	12345	12345	Ahmed	✗Reject: Password too short
TC06	Abdullah	abdullah123@gmail.com	123456	65785	Abdullah	✗Reject: Passwords don't match
TC07	AbdullahQasim	qasim89@gmail.com	123456	123456	(blank)	✗Reject: Name blank (if required)
TC08	Minahil	mm@gmail.com	123456	123456	Minahil	✓Accept

Login Page:

Field	Range	Valid Range Description	Boundary Values Tested	Expected Outcome
Username	2–25 characters	Must be between 2 and 25 characters	2, 3, 24, 25, 17	1 & 26: ✗ Fail 2–25: ✓ Pass
Password	6–12 characters	Must be between 6 and 12 characters	6, 7, 11, 12, 9	5 & 13: ✗ Fail 6–12: ✓ Pass

Test Case ID	UserName	Password	Expected Outcome
TC01	(blank)	123456	✗Reject: Username blank
TC02	a	123456	✗Reject: Username too short (<3 characters)
TC03	minahilazeemaze emminahil	123456	✗Reject: Username too long
TC04	MinahilAzeem	(blank)	✗Reject: Password blank
TC05	AhmedBaig	12345	✗Reject: Password too short
TC06	Abdullah	12345688888888888888	✗Reject: Password too long (>16 characters)
TC07	AbdullahQasim	123456	✓Accept

TC08	Minahil	(wrong password)	✗Reject: Invalid credentials (mismatch)
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Lessons Learned

Throughout the development of *Sheher-e-Sukhan*, our team encountered a variety of technical, collaborative, and creative challenges that helped us grow both as developers and as team members. Key lessons include:

1. Effective Team Coordination Is Crucial

Clear role assignment and regular check-ins helped prevent confusion and kept the project on track. We learned the value of consistent communication using tools like Trello, WhatsApp, and GitHub.

2. Planning Saves Time

Taking time to design the database schema, user flow, and core features before writing code reduced rework and made development smoother.

3. Divide and Conquer Works Best

Splitting the project into sprints and assigning features to individuals based on strengths (e.g., frontend, backend, data handling) led to more focused and efficient progress.

4. Version Control Is a Lifesaver

Using GitHub for code management taught us how essential version control is for collaboration and tracking progress, especially when multiple people are working simultaneously.

5. User Feedback Improves Quality

Informal testing and feedback from peers gave us insights we hadn't considered during development—especially around UI/UX and feature accessibility.

6. **Expect the Unexpected**

We faced unexpected issues like bugs during integration, merging conflicts, and misunderstood requirements. These taught us to stay flexible and resilient.

7. **Documentation Matters**

Maintaining clear documentation for the database, APIs, and project features not only helped us stay organized but will also support future enhancements or handovers.
